Momentum analysis: put it together

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a model to compare a deflection option and its limits.

SUCCESS CRITERIA - I can...

1. Compute the required change before evaluating an option.
2. Keep impulse, velocity change and displacement distinct.
3. Communicate assumptions with the numerical result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to compare your model or result. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

A system’s total momentum is 15 kg·m/s west and 21 kg·m/s south. Its combined mass is 3 kg.
Find final velocity components.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: shift = Δv × time; J = MΔv.
Fictional straight-line planning model. M is target mass; J is required impulse. This is not a real encounter forecast.

Worked reasoning: A 1 kg cart at +6 m/s sticks to a 2 kg cart at rest. Final velocity is +2 m/s. The moving cart changes momentum by -4 and the other by +4 kg·m/s.

Your turn: Draw both momentum changes and show how their sum matches the system change.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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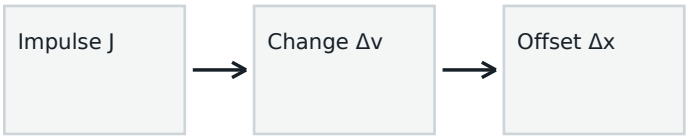
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Organize this chain with units: impulse → velocity change → sideways shift. Which step uses target mass?

Sentence starter: I notice … . This supports … because … .

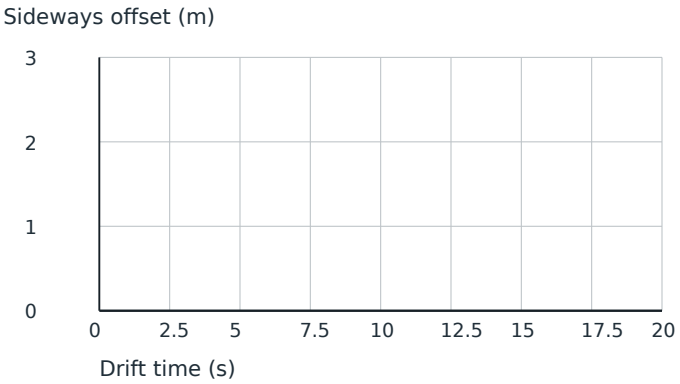


Marzano check - circle: ① ② ③

04 Apply the physics

In a simplified deflection simulation, a 100 kg target needs a 2 m sideways offset after a 20 s drift. Assume the added sideways velocity stays constant after a brief push. **Find** the required velocity change, then the impulse.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

If the target mass estimate is too low, how would a fixed delivered impulse's velocity change compare with the prediction?

Word help: estimate = find an approximate value

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 4 kg cart moving at 6 m/s stops in 3 s.
Find momentum-change magnitude and average stopping-force magnitude.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, including direction

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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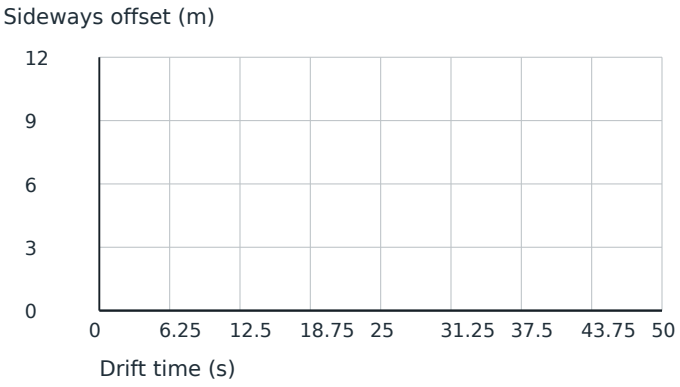
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A brief push adds 0.2 m/s sideways to a probe’s velocity. The probe then drifts for 50 s. Complete the position-time line, find its sideways offset and state one assumption.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. The same pair now receives an external impulse of 3 N·s right during contact.
Find final total momentum and shared velocity. Organize the known quantities on the before-and-after model.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3